TrustBridge – English table set (working preview)

November 1, 2026

Numbers from docs/12 (2026-10-12 consolidation). Each `tab*.tex` is a standalone `tabular` block for `\input` into the final paper.

Table 1: Main results on IXI (self-trained 128 × 128 models, 60 test slices, official mean-norm PSNR/SSIM protocol). AUROC: low-trust pixels localizing the top-error decile; chance level 0.5.

	Translation quality			Trust localization				
Direction	PSNR (dB)	SSIM (%)	NCC	Spearman	Pearson	AUROC (fused)	AUROC (RCRF)	AUROC (grad.)
T2→T1	20.04±1.18	71.33	0.887	−0.806	−0.493	**0.819±0.064**	0.800	0.448
T1→T2	20.68±1.13	80.59	0.908	−0.902	−0.592	**0.902±0.026**	0.887	0.777

Table 2: Seven-row ablation: ΔPSNR (dB) after dropping the least-trusted 20% of pixels (60 test slices per direction).

Scheme	T2→T1	T1→T2	Notes
Random	−0.066	−0.011	falsifies the protocol
Gradient/edge	+0.018	+1.455	unsupervised prior baseline
Input-difference†	+2.022	+0.385	GT-leaking reference
RCRF only	+1.911	+4.660	contribution (1) alone
Ensemble only	+1.932	+4.934	contribution (2) alone
Fused (0.6/0.4)	**+2.136**	**+5.187**	$\geq$ either single signal
Oracle (uses GT)	+10.132	+15.031	upper-bound reference

Table 3: Fusion-weight sensitivity (w_1 = RCRF weight) and step skipping (IXI test set, 60 slices, ΔPSNR@20% in dB).

w_1 ($w_2 = 1 - w_1$)	T2→T1	T1→T2
1.0 (RCRF only)	2.403	4.728
0.7	2.712	5.248
0.5	2.811	5.439
0.3	**2.841**	**5.472**
0.0 (ensemble only)	2.448	4.900
Step skipping, skip=2 (fusion weights 0.6/0.4):		
skip=1 (per-step, 40 fwd passes)	2.771 (3.92 s/slice)	5.354 (3.10 s/slice)
skip=2 (24 fwd passes, −39% time)	2.859 (2.38 s/slice)	5.494 (1.89 s/slice)

Table 4: Cross-dataset results. CT slices come from two centers with different weightings (A: T1w, C: T2w); weights are center-matched.

Dataset	Direction	Weights	n	PSNR (dB)	NCC	Spearman	AUROC (fused)
BraTS2018	FLAIR→T2	official	60	**24.06±1.62**	0.944	−0.897	**0.932±0.029**
Pelvis CT, ctr. A (T1w)	T1→CT	official	80	21.24±2.42	0.936	−0.673	0.915±0.031
Pelvis CT, ctr. C (T2w)	T2→CT	official	80	19.04±0.73	0.893	−0.626	0.705±0.033
Pelvis CT, ctr. A (T1w)	T1→CT	self-tr.	80	**24.36±2.92**	0.962	−0.808	0.869±0.015
Pelvis CT, ctr. C (T2w)	T2→CT	self-tr.	80	**23.13±0.83**	0.961	−0.766	0.726±0.019

Table 5: Lesion-indication analysis on BraTS2018 (399 slices, unseen by the translated model).

Metric	Value
Trust map indicating tumor region, AUROC	**0.871**
True-error map indicating tumor region, AUROC (reference)	0.925
Mean abs. error inside vs. outside tumor	1.54×
Mean trust inside vs. outside tumor	0.795 vs. 0.725
Wilcoxon signed-rank test	$p < 10^{-300}$

Table 6: Calibration analysis: mean absolute error of the least- vs. most-trusted decile (80 slices per direction).

Direction	Monotonicity	Error separation	Spearman (trust, error)
T2→T1	0.889	35.3× (0.72 → 0.020)	−0.806
T1→T2	0.889	73.2× (1.07 → 0.015)	−0.903